RBC CAPITAL MARKETS · DIGITAL ASSET TECHNOLOGY · EXECUTIVE SUMMARY

Digital Asset Subledgers

What Line of Business subledgers, sitting between 24/7 chains and the General Ledger, are worth to Finance, and how they connect the Lines of Business without a shared execution hub.

Head of Digital Asset Technology, RBC Capital Markets · October 28, 2026 · Summary of the full position paper

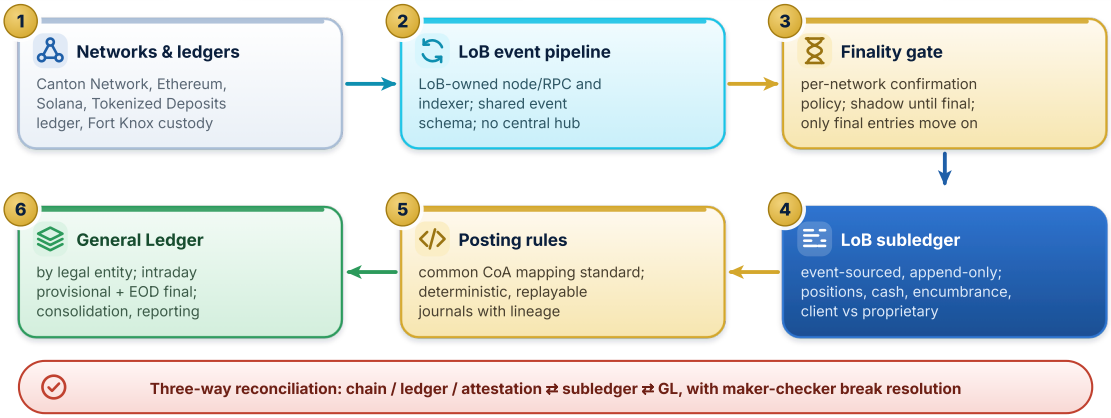
The position in one line: real time belongs in the Line of Business subledgers and the Capital Markets intraday core, not in the GL. Each subledger is an event-sourced, reconciled, real-time operational book that posts summarized, rule-driven journals to the GL on defined windows. The GL stays the aggregate book of record; a chain is not a book of record, though a token ledger can be the golden source for holdings. Some Lines of Business may share a subledger, partitioned by legal entity, Line of Business, book and product, while execution, signing, policy and GL mapping stay with each business.

7
Lines of Business, each with a dedicated or shared subledger

1
common central path to Fort Knox and the Tokenized Deposits ledger

3-way
reconciliation: chain or ledger ⇄ subledger ⇄ GL

24/7 → close
per-entity cut-off turns continuous activity into closed days



The subledger journey. Networks → Line of Business pipeline → finality gate → event-sourced subledger → posting rules → GL, with three-way reconciliation across all three records.

General Ledger

Aggregate book of record by legal entity, account and period. Built for statements, capital and liquidity returns and ICFR, not for wallet movements.

Line of Business subledger

Positions, cash, encumbrance, pending vs settled, fees and gas, client vs proprietary. Produces journals and the audit drill-down.

Chain or token ledger

Token state and transfers. Internal ledgers are authoritative; external networks are mirrored and attested. The subledger handles network finality internally.

VALUE TO FINANCE

Where the value shows up, stated qualitatively

Reconciliation — RECON
Three-way and continuous, with a break taxonomy, owners, ageing and maker-checker. Breaks surface near when they occur, not at month-end.

Intraday visibility — TREASURY
Shadow and authoritative balances per position. Intraday view of tokenized-deposit liabilities and Capital Markets settlement and financing.

Close acceleration — CLOSE
Accounting date derived from finality time against a per-entity cut-off; closed, immutable event sets; provisional entries always true up.

GL posting — GL
One CoA mapping standard; no GL account per token or network; deterministic, replayable journals. Treatment follows the instrument, not the technology.

Controls and audit trail — ICFR
Hash-chained event log; every journal line traceable to events, rule version and Watchtower decision ID. ICFR evidence by design.

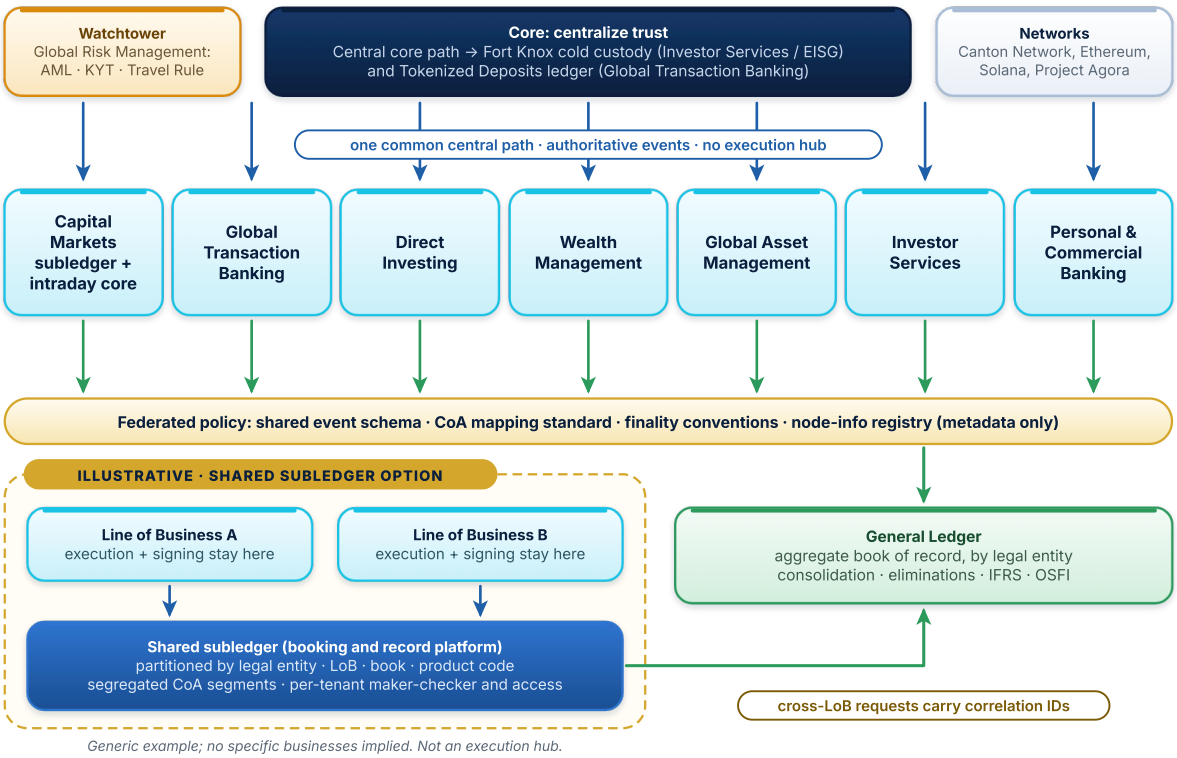
Capital and liquidity — PRUDENTIAL
Basel / OSFI classification attributes per position, versioned. OSFI's 2027 crypto-asset guideline is effective 1 November 2026 for October year-end banks VERIFY.

Cost and efficiency — COST
Standards built once and reused by seven Lines of Business; no bespoke Fort Knox or Tokenized Deposits integrations; replay instead of manual rework.

Intercompany — CONSOLIDATION
Shared correlation keys (fulfillment ID, settlement ID) make matching and IFRS 10 elimination mechanical.

No benefit figures are claimed. Finance would baseline reconciliation effort, aged breaks, days to close, manual journal share, intraday visibility lag and run cost per Line of Business before any number is quoted [baselines TBD].

Connecting the Lines of Business: centralize trust, federate policy, decentralize execution



Generic example; no specific businesses implied. Not an execution hub.

Core-Edge. One central path to the Core services; shared standards; each Line of Business executes and keeps its own book; GL by legal entity. The inset is an illustrative example of two Lines of Business sharing one partitioned subledger, not a statement of which businesses share. No shared execution hub, central Kafka hub or central hot KMS.

OPERATING MODEL, RISKS AND NEXT STEPS

How it runs and what it needs

Role	Owns
Core	Central core path; Fort Knox cold custody (Investor Services / EISG); Tokenized Deposits ledger; completeness guarantees
Shared standards	Event schema, CoA mapping standard, finality conventions, node-info registry (metadata only), conformance tests
Each Line of Business	Pipeline, node / RPC, subledger (dedicated or a partition of a shared one) with its own policy, approvals and GL mapping, reconciliation instance, break ownership, hot / warm signing for external networks
Finance	Posting rules and versions, journal approval, cut-offs, close, consolidation, accounting policy
Global Risk Management Watchtower	Pre-movement gates (AML / KYT, sanctions, Travel Rule, limits), monitoring, key-event escalation

Posting on non-final data

The subledger handles network finality internally; Finance and the GL only ever receive final, authoritative entries.

Client entitlement exceeds custody

Continuous reconciliation per custody location; key event; segregation at wallet level.

Missed events or central-path outage

Offset continuity checks; independent attestation node; pending-not-posted until completeness is proven.

Classification drift

Versioned, position-level attributes so Basel or OSFI changes are reference-data changes.

DEDICATED OR SHARED SUBLEDGER

Sharing fits similar products, the same legal entity or booking model, or a shared cash leg (e.g. Global Transaction/Cash Management for other businesses). It is a shared **booking and record** platform, not an execution hub: partitions by legal entity, Line of Business, book and product code, segregated CoA segments and per-tenant maker-checker. Value: fewer reconciliations, reuse, lower run cost [baseline TBD]. Risks: blast radius, change coordination, entitlement leakage. Each business keeps its own policy, approvals and GL mapping.

PHASE 0
Decide and baseline
Legal entities, policy elections, registers; Finance baselines; publish shared standards.

PHASE 1
Internal ledgers
Global Transaction Banking subledger on the Tokenized Deposits ledger; ledger = subledger = GL control.

PHASE 2
Capital Markets
Subledger downstream of the intraday core; DvP with a tokenized-deposit cash leg; three-way reconciliation live.

PHASE 3
Extend by conformance
Direct Investing, Investor Services, then Wealth Management, Global Asset Management, Personal & Commercial Banking.

Decisions requested

- **Endorse the principle:** real time in subledgers and the intraday core; summarized, rule-driven journals to the GL.
- **Mandate the shared standards** as a condition for any Line of Business going live.
- **Name Finance owners** for posting rules, cut-offs and the benefit baseline.
- **Agree Phase 1 scope** with Global Transaction Banking on the Tokenized Deposits ledger.

Prepared by the Head of Digital Asset Technology, RBC Capital Markets. Summary of "Digital Asset Subledgers by Line of Business: value to Finance" (position paper, October 28, 2026), expanded from the source paper on GL integration and real-time synchronization. Items tagged VERIFY need confirmation with Finance Policy, Legal, Regulatory Reporting or the network operator. Not accounting, legal or regulatory advice.